

Johnson C. Smith University
Calculus II | MTH 232 | Section C | Fall 2012

“Every act of conscious learning requires the willingness to suffer an injury to one’s self-esteem. That is why young children, before they are aware of their own self-importance, learn so easily; and why older persons, especially if vain or important, cannot learn at all.”

-Thomas Szasz, author, professor of psychiatry (1920-)

Instructor Information

Name: Dr. James Ashe

Office location: 307 Davis Hall (back); office hours will also be held in the Math Lab, EDU 115

Email: jashe@jcsu.edu

Office phone: (704) 378-1037

Office hours: MWF 10–11am; MW 1–4pm; Tu 12pm–2pm; F 1pm–1:30pm, by appointment, or whenever I’m in my office.

Course Description

This is a **3 credit hour** course; the second of a sequence of four calculus courses which are offered for students from the sciences, social sciences and engineering. Topics covered: Anti-derivatives, Fundamental Theorem of Calculus and definite integrals, Applications of the definite integral to area, volume, force and work, and arc length. Transcendental functions. Techniques of integration. Prerequisite: MTH 231 (Calc I).

Instructional methods will primarily consist of lectures and in-class discussions, however other methods may be adopted according to the particular needs of the class.

Course Materials

Textbook: *Calculus, Briggs and Cochran. ISBN: 978-0321336118*

Calculator: TI-83, TI-83+, or TI-84+

Students should bring their calculator and textbook to class.

Grades

Grades will be taken from homework, quizzes, and Exams as follows:

Grading scale: $A \geq 90 > B \geq 80 > C \geq 70 > D \geq 60 > F$

Evaluation: Quizzes = 20%; Exams 1, 2, 3, and 4 = 15% each; Final Exam = 20%.

Quizzes: Quizzes will consist of any graded assignment which is not an exam. This will include traditional quizzes, take home assignments, and in-class projects. The lowest quiz grade will be dropped.

Final Exam: The final Exam will be a comprehensive in-class exam; see the exam schedule:

jcsu.edu/happenings/exam-schedule. If beneficial, the final exam score will replace the lowest exam score.

Class Policies

JCSU policies regarding attendance, academic integrity, grades, and change of enrollment will be enforced. It is the student’s responsibility to be aware of these policies and of related deadlines as well as any announcements or syllabus adjustments (written or oral) made in class. Please see the student handbook for details, jcsu.edu/docs/handbook.pdf.

Attendance: Class attendance is required for all JCSU students. Each student is allowed as many hours of absence per term as credit hours(s) received (not to exceed 4) for the class. Students who exceed the maximum number of absences may receive a failing grade for the course.

Students who may miss classes while representing the University in an official capacity are exempt from regulations governing absences. However, absence from class for official University business does not relieve the student from responsibility for any class assignments that may be missed during the period of absence.

Missed quizzes: There will be no make-ups for in-class quizzes unless the class was missed for a University sponsored event, and advance notice of the absence was given by the student. In this case the assignment will be excused or a make-up will be given depending on circumstances. Late take-home quizzes will have 25% deducted for each day it is late.

Missed exams: If a student misses an exam for a verifiable excuse, they should send me an e-mail *as soon as possible* so that a make-up can be arranged. Make-up exams may be significantly more difficult. If you will be gone for any reason, it will be *your* responsibility to inform me at least a week in advance to schedule to take the exam early.

Student support services: If you need course adaptations or accommodations for a documented disability please contact the Office of Disability Services.

jcsu.edu/directory/federal-requirements/general-institutional-information/jcsu-disability-services,
phone: (704) 398-1116.

The Office of Counseling Services provides confidential intakes and assessments, personal counseling and referrals to appropriate resources to all enrolled students on a walk-in or appointment basis.

jcsu.edu/admissions/future_students/meet_your_counselor, phone (704) 378-1044.

Course Content

This is an outline of topics that will be covered in this course.

Chapter 4: Applications of the Derivative

4.8 Antiderivatives

Chapter 5: Integration

5.1 Approximating areas under curves

5.2 Definite Integrals

5.3 Fundamental Theorem of Calculus

5.4 Working with Integrals

5.5 Substitution Rule

Chapter 6: Applications of Integration

6.1 Velocity and Net Change

6.2 Regions Between Curves

6.3 Volume by Slicing

6.5 Length of Curves

Chapters 7: Logarithmic and Exponential Functions

7.1 Inverse Functions

7.2 The Natural Logarithmic and Exponential Functions

7.3 Logarithmic and Exponential Functions with other Bases

Chapter 8: Integration Techniques

8.1 Integration by parts

8.2 Trigonometric Integrals

8.3 Trigonometric Substitution

8.4 Partial Fractions

8.7 Improper Integrals

8.8 Introduction to Differential Equations

Important Dates

September 21 th	Last day to Add/Drop
October 15 th	Exam 1
November 2 nd	Exam 2
November 28 th	Exam 3
January 7 th	Exam 4
January 15 th –19 th	Final Exam

With exception of the final, the above exam dates are subject to change. JCSU's Fall 2012 academic calendar can be found here: jcsu.edu/happenings/academic-calendar, and the exam schedule can be found here: jcsu.edu/happenings/exam-schedule